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COLLEGE OF ENGINEERING AND TECHNOLOGY
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Department of Computer Science & Engineering

Topic: Digital Twin-Based Predictive Maintenance for CNC Machines Using Vibration Sensor Fusion and LSTM Anomaly Forecasting at the Edge

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ABSTRACT

CNC machines are widely used in manufacturing industries for precise machining operations. Unexpected failures in CNC machines can cause production delays, maintenance cost, and downtime. Traditional maintenance methods are based on periodic inspection and cannot predict sudden faults effectively.

This project proposes a Digital Twin-based predictive maintenance system using vibration sensor fusion and LSTM anomaly forecasting at the edge. Simulated vibration data is collected, processed locally, and analyzed using an LSTM model to predict abnormal machine behavior. The system provides alerts and dashboard visualization for efficient maintenance planning.

INTRODUCTION

Industry 4.0 focuses on smart manufacturing using IoT, AI, and automation. CNC machines are important assets in production industries. Continuous operation may lead to bearing wear, spindle imbalance, motor vibration, and tool faults.

Digital Twin technology creates a virtual representation of physical machines. By combining vibration monitoring and machine learning, predictive maintenance can reduce unexpected failures and improve machine life.

PROBLEM DEFINITION

Current CNC maintenance systems face several problems:

- Unexpected machine breakdown
- High maintenance cost
- Production downtime
- Manual inspection delays
- No early fault prediction
- Poor maintenance planning

OBJECTIVES

- To develop a Digital Twin model for real-time CNC machine monitoring and analysis.
- To collect and analyze vibration data for early fault detection in machine components.
- To apply sensor fusion techniques by combining multiple vibration signals accurately.
- To implement LSTM deep learning model for anomaly forecasting and prediction.
- To reduce unexpected downtime, repair cost, and maintenance expenses.
- To provide an interactive dashboard for machine health monitoring and alerts.
- To improve production efficiency, reliability, and machine lifespan.

LITERATURE SURVEY

Several research works have been conducted in machine monitoring and predictive maintenance systems.

1. **IoT based Machine Monitoring Systems**

   Used sensors and cloud platforms for monitoring industrial machines.

2. **AI based Predictive Maintenance**

   Machine learning models used to predict machine failures based on sensor data.

3. **Edge Computing for Industrial Systems**

   Edge devices process data locally to reduce latency and improve real-time decision making.

4. **Time Series Data Monitoring Platforms**

   Tools like InfluxDB and Grafana are widely used for industrial monitoring dashboards.

EXISTING SYSTEM

The existing CNC machine maintenance systems mainly use the following methods:

1.Manual Maintenance Inspection

Operators check machine condition manually based on sound, vibration, and performance.

2.Preventive Maintenance Scheduling

Maintenance is performed at fixed time intervals regardless of machine health.

3.Threshold-Based Alarm Systems

Alerts are generated only when sensor values cross predefined limits.

4.Cloud-Only Monitoring Platforms

Machine data is sent to cloud servers for monitoring and analysis.

5.Reactive Maintenance Approach

Repairs are carried out only after machine failure occurs.

6.Basic Historical Record Analysis

Past maintenance records are used for decision making.

DEMIRITS

- Existing systems cannot predict sudden machine failures accurately in advance.
- High downtime occurs during unexpected machine breakdown conditions.
- Requires frequent human monitoring and manual inspection process.
- Fault identification is delayed, causing production interruptions.
- Maintenance planning is less efficient and increases overall cost.
- No intelligent AI-based forecasting for machine health analysis.
- Traditional methods reduce productivity and machine utilization.

PROPOSED SYSTEM

The proposed system introduces a smart predictive maintenance platform using Digital Twin technology. A virtual model of the CNC machine is created to monitor real-time health conditions. Simulated vibration data from spindle, bearing, and motor components are continuously generated.

Sensor fusion combines multiple vibration channels into a single meaningful dataset. This fused data is transmitted to the edge processing module. The LSTM neural network model analyzes historical patterns and predicts future anomalies. If unusual vibration behavior is detected, alerts are sent to the maintenance dashboard.

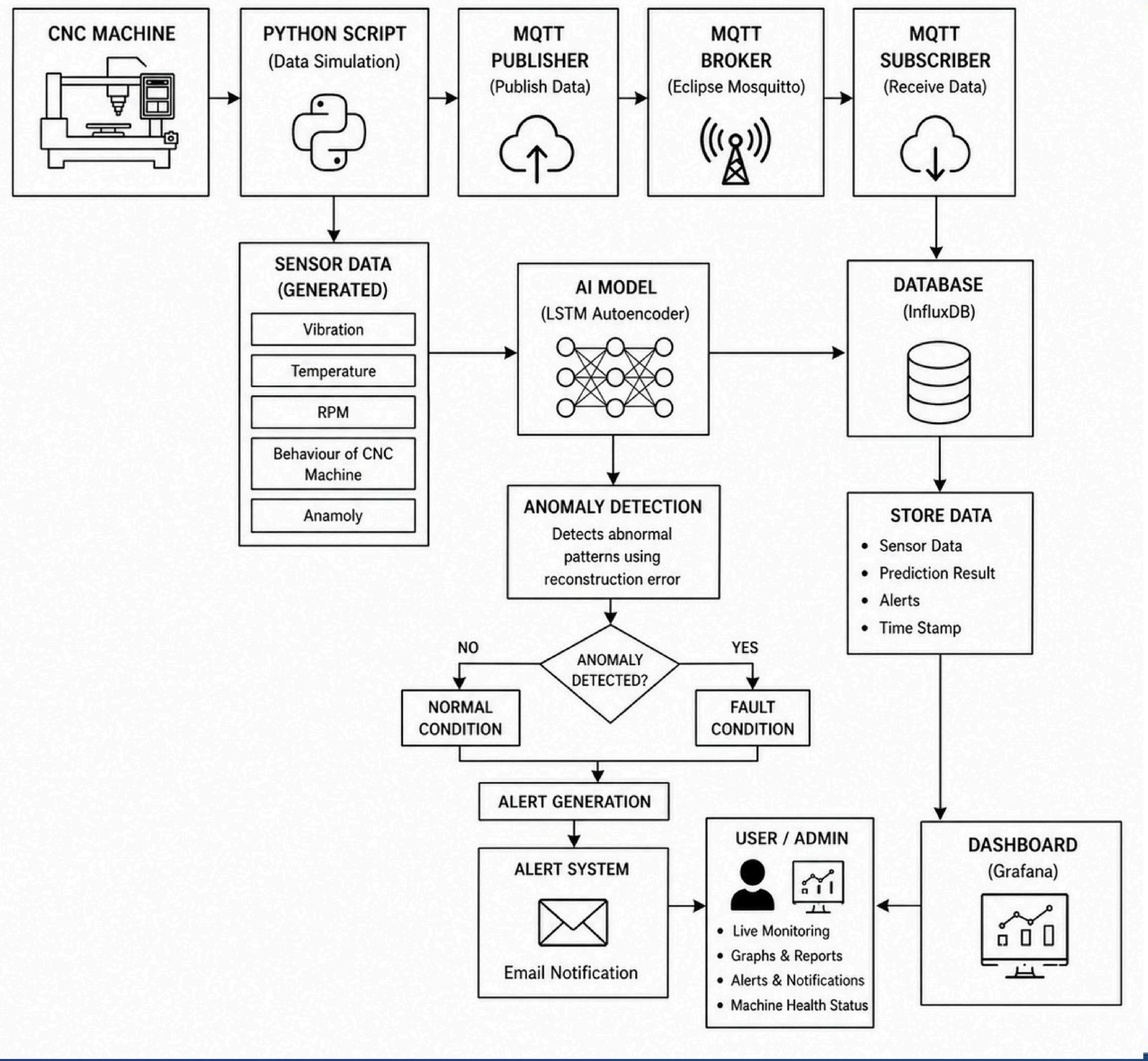
The dashboard displays graphs, health score, anomaly trends, and warning messages. Since prediction happens at the edge, response time is faster and cloud dependency is reduced.

ADVANTAGES

The proposed system offers several benefits:

- Real-time machine condition monitoring
- Early fault detection before breakdown
- Reduced downtime and production loss
- Lower maintenance and repair cost
- Better machine utilization
- Improved safety and reliability
- Smart dashboard visualization
- Fast local prediction using edge computing
- Scalable architecture for multiple machines
- Easy integration with future IoT sensors

SYSTEM ARCHITECTURE



TECHNIQUES USED

1. **Python** Core programming language for data generation, preprocessing, MQTT communication and AI model integration.
2. **PyTorch** (LSTM Autoencoder) Deep learning framework used to implement LSTM Autoencoder model for anomaly detection based on reconstruction error.
3. **MQTT** Protocol Lightweight publish-subscribe communication protocol for real-time data transmission between edge devices and processing system.
4. **InfluxDB** Time-series database for storing continuous streams of sensor data, prediction results and anomaly status with timestamps.
5. **Grafana** Interactive dashboard tool for real-time visualization of machine parameters, anomaly detection results and alert generation.
6. **Docker** Containerization platform used to deploy and manage MQTT broker, InfluxDB and Grafana components for easy scalability and portability.

ALGORITHM

1. Start
2. Collect vibration sensor data
3. Preprocess data
4. Send through MQTT
5. Store in database
6. Load trained LSTM model
7. Predict next vibration values
8. Compare with threshold
9. If anomaly detected → Alert
10. Update dashboard
11. Stop

MECHANISM

The system starts by collecting or simulating vibration values from CNC components. Data preprocessing removes noise and standardizes values. Sensor fusion combines multiple vibration streams into a single reliable signal.

The fused data is passed to the LSTM model deployed at the edge device. The model predicts future vibration trends based on historical patterns. If deviation exceeds the normal operating range, the system classifies it as anomaly.

The anomaly status, machine health score, and graphs are displayed in the Digital Twin dashboard. Maintenance staff can then take preventive action before failure occurs.

MODULES OF THE SYSTEM

1.**Sensor DataCollection Module**

Collects real-time machine d ata such as temperature, vibration, and RPM.

2.**MQTT Communication Module**

Transfers sensor data from edge devices to the processing system.

3.**Data Processing & AI Module**

Processes incoming sensor data and detects an omalies using AI algorithms.

4.**Data Storage Module**

Stores time-series m achine data in *InfluxDB database*.

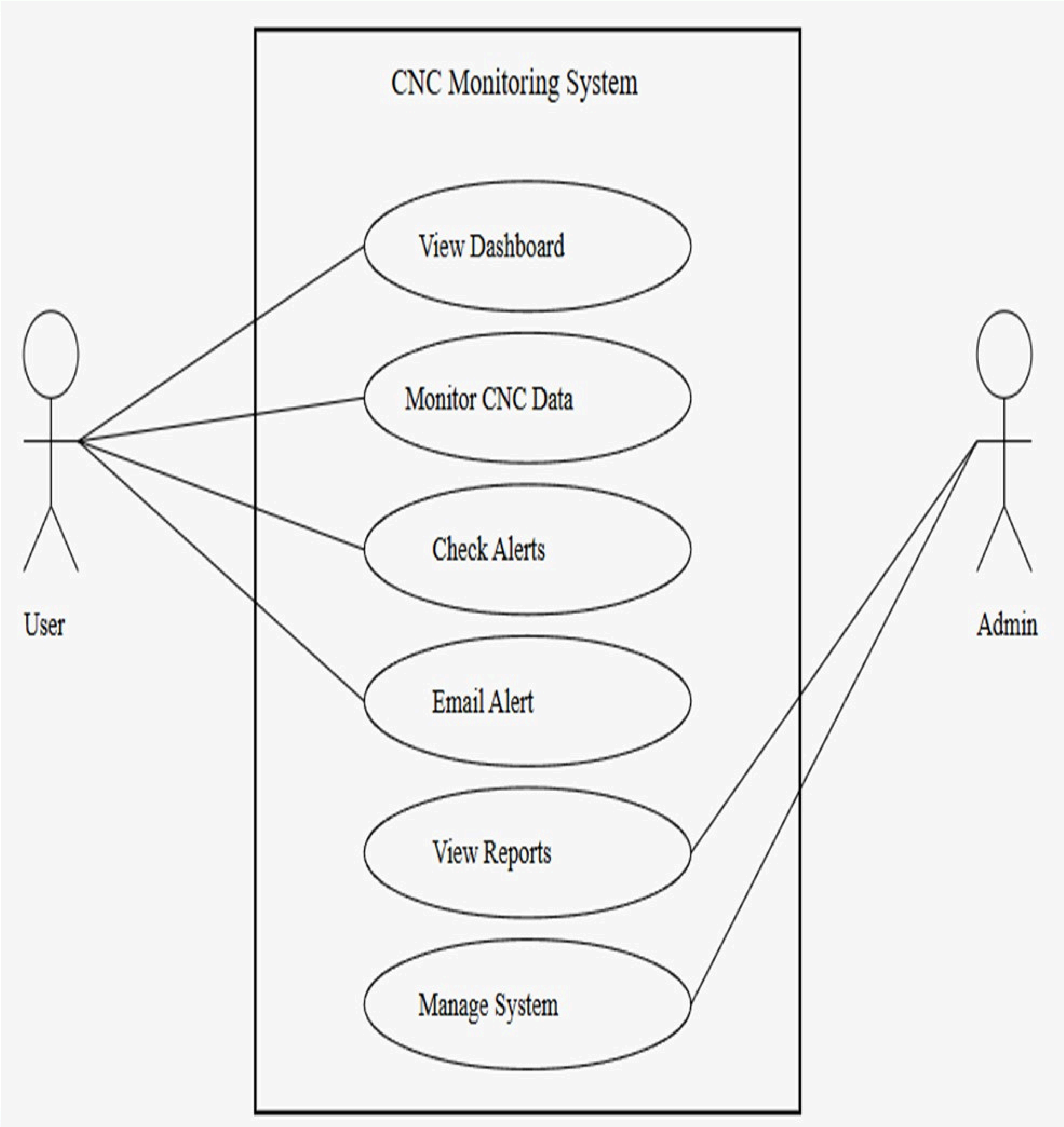
5.**Visualization Module**

Displays real-time machine performance using *Grafana dashboard*.

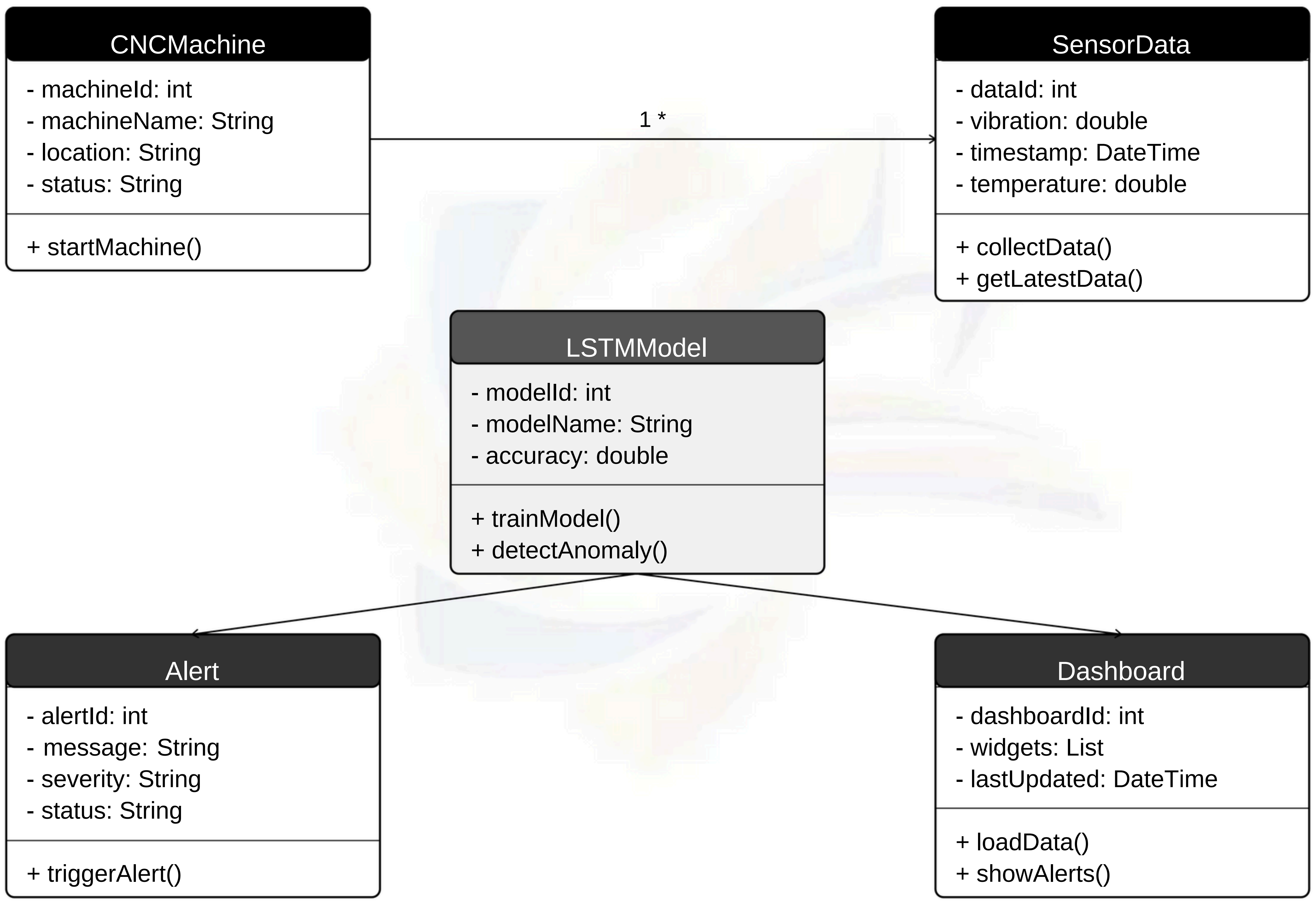
6.**Alert & Notification Module**

Sends alerts via email when abnormal conditions are detected.

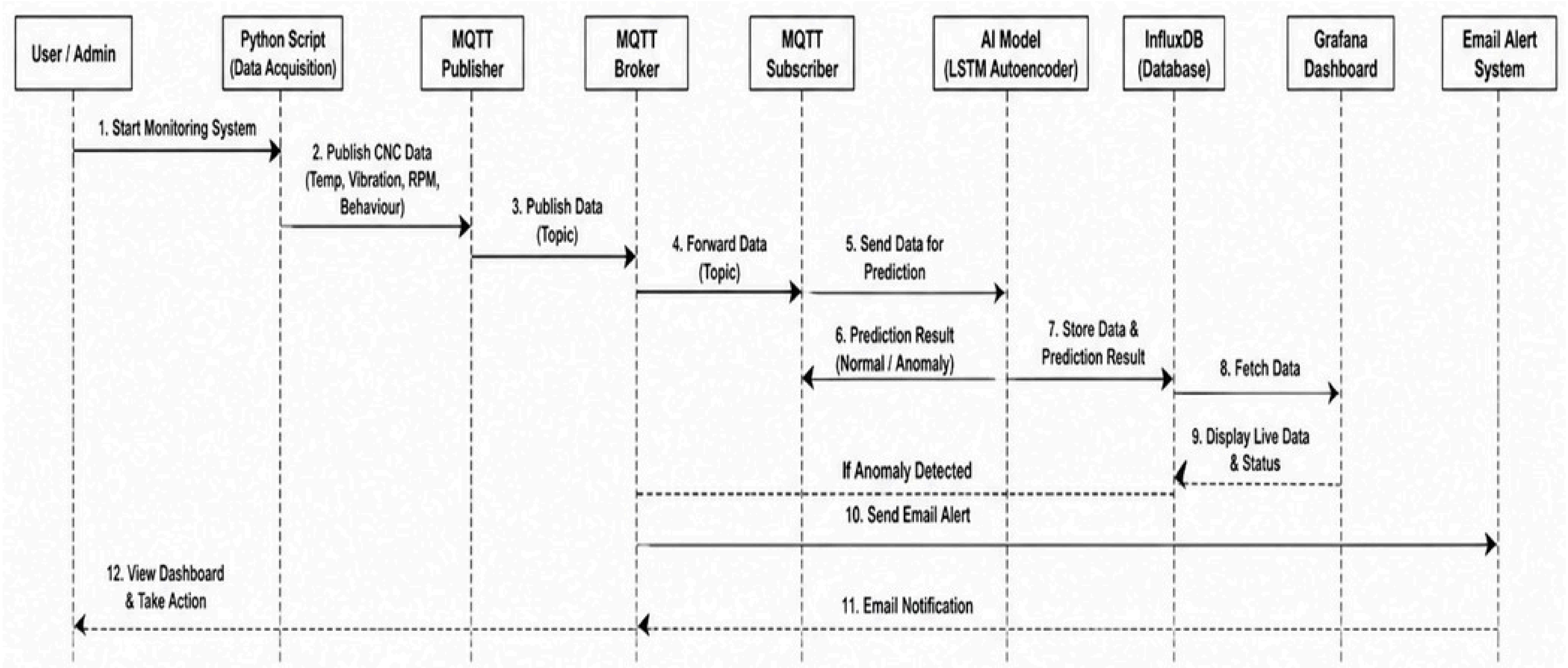
UML DIAGRAM (Use Case Diagram)



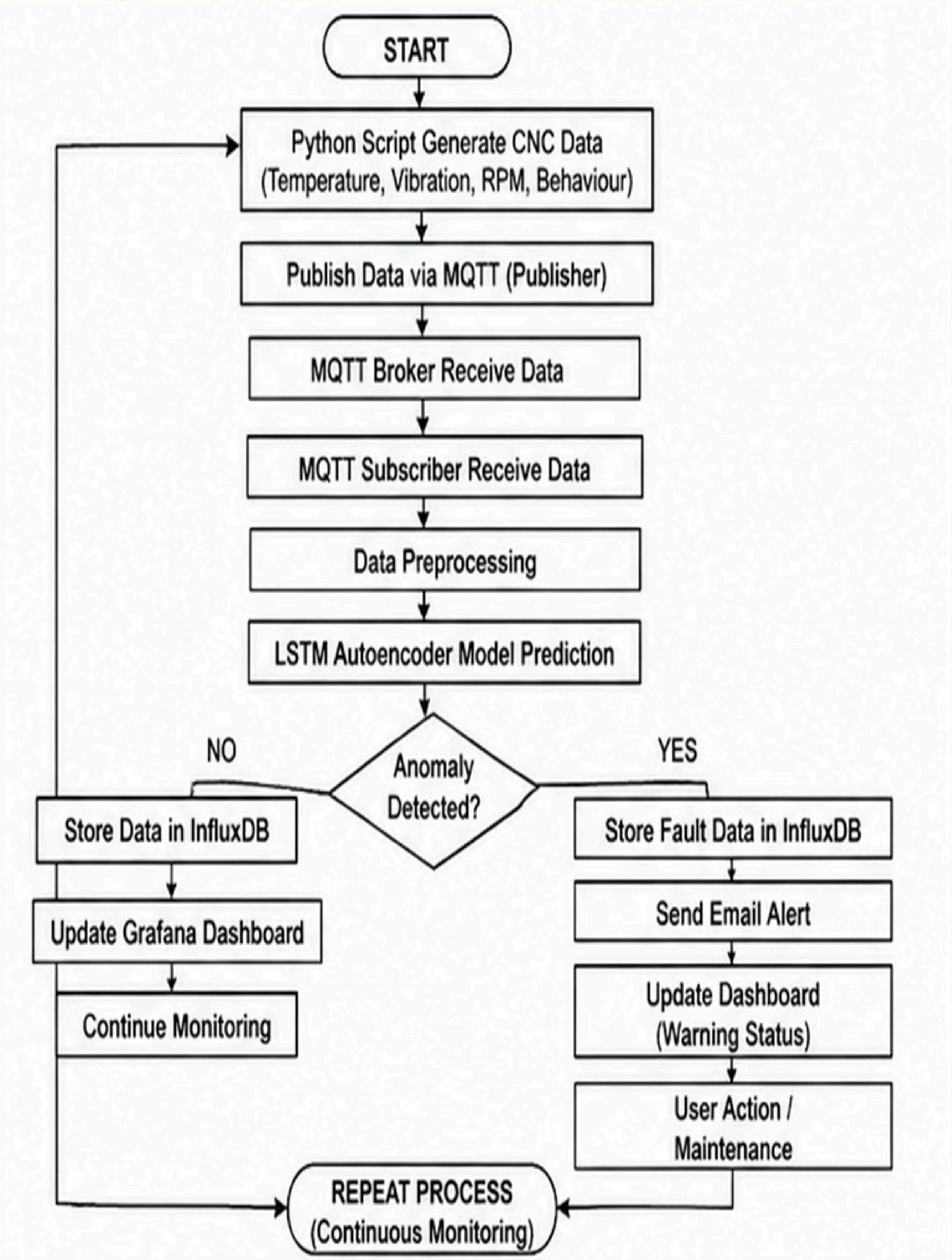
UML DIAGRAM (Class Diagram)



UML DIAGRAM (Sequence Diagram)



UML DIAGRAM (Activity Diagram)



RESULTS AND OUTPUT


```
Sent MQTT: {"temperature": 70.41, "rpm": 4674, "vibration": 6.32, "anomaly": 0}
Sent MQTT: {"temperature": 79.45, "rpm": 3286, "vibration": 8.63, "anomaly": 1}
Sent MQTT: {"temperature": 70.83, "rpm": 3478, "vibration": 6.78, "anomaly": 0}
Sent MQTT: {"temperature": 79.06, "rpm": 1058, "vibration": 6.89, "anomaly": 0}
Sent MQTT: {"temperature": 82.6, "rpm": 1955, "vibration": 7.71, "anomaly": 0}
Sent MQTT: {"temperature": 57.88, "rpm": 3159, "vibration": 6.99, "anomaly": 0}
Sent MQTT: {"temperature": 74.89, "rpm": 4527, "vibration": 9.66, "anomaly": 1}
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Sent MQTT: {"temperature": 86.04, "rpm": 3825, "vibration": 9.84, "anomaly": 1}
Sent MQTT: {"temperature": 75.24, "rpm": 4897, "vibration": 7.05, "anomaly": 0}
```
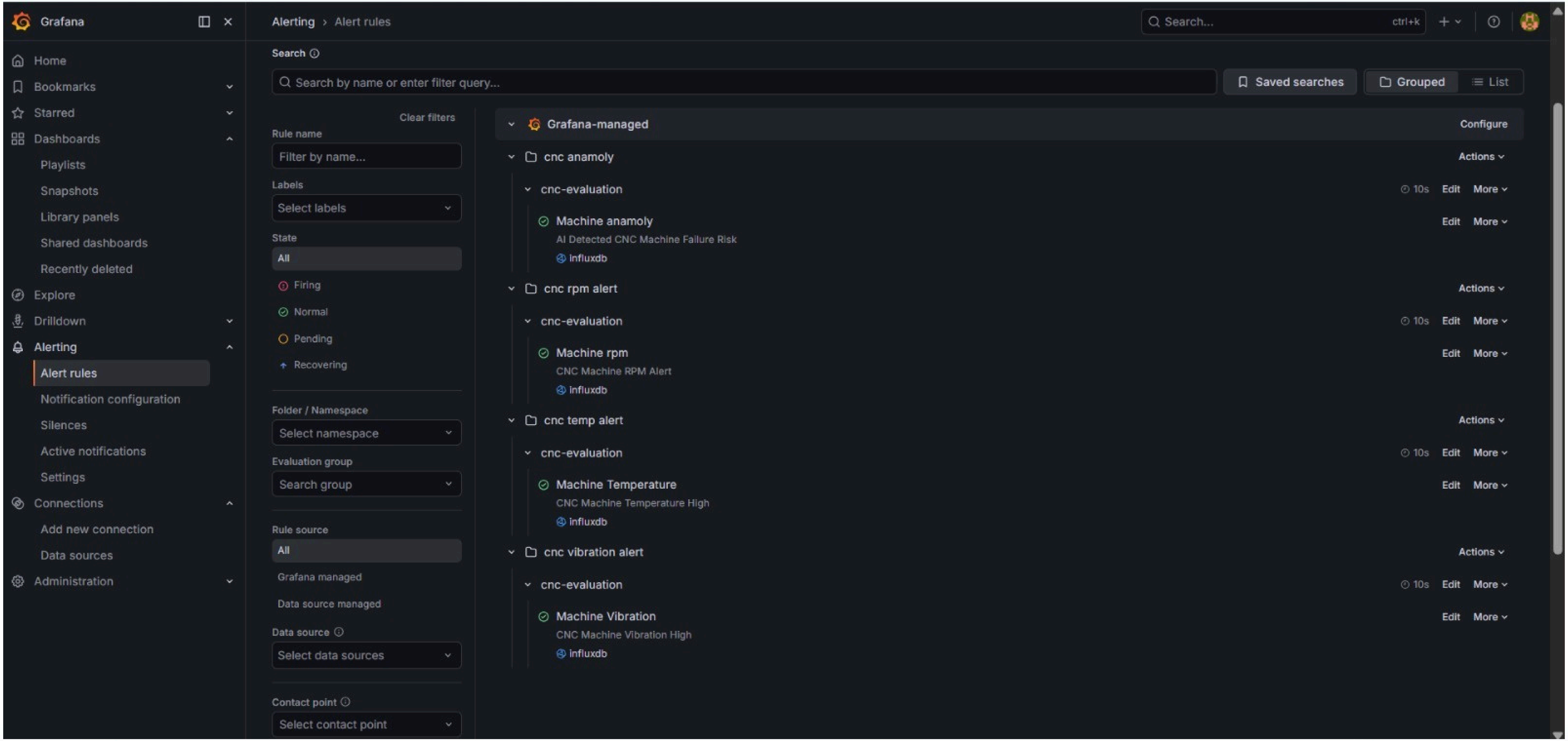
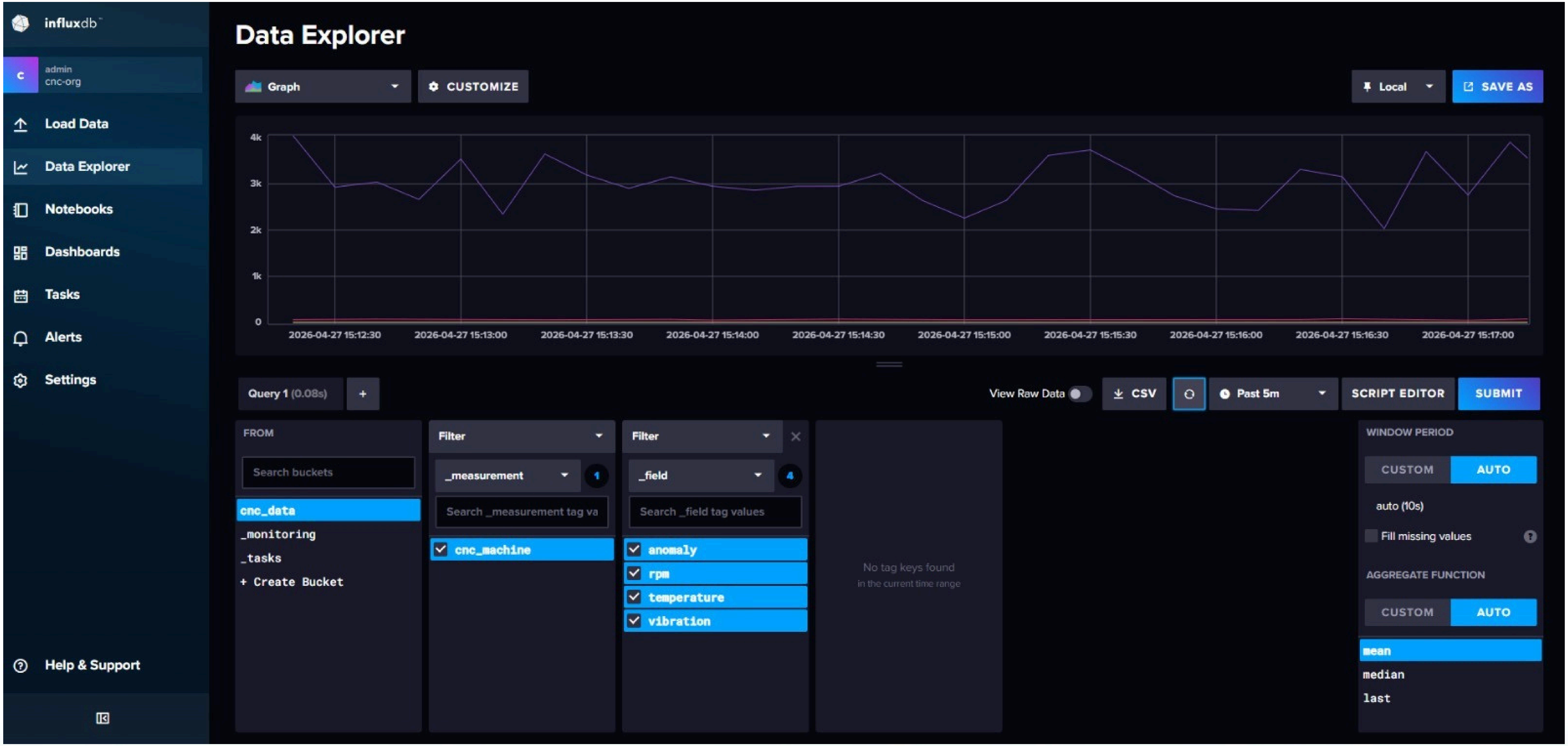

RESULTS AND OUTPUT

```
RPM: 1497.0
Vibration: 8.94
STATUS: FAILURE RISK

EDGE AI DETECTOR
------------------------
Temp: 46.46
RPM: 4038.0
Vibration: 9.14
STATUS: FAILURE RISK

EDGE AI DETECTOR
------------------------
Temp: 48.04
RPM: 3586.0
Vibration: 9.36
STATUS: FAILURE RISK

EDGE AI DETECTOR
------------------------
Temp: 50.73
RPM: 1598.0
Vibration: 8.02
STATUS: WARNING

EDGE AI DETECTOR
------------------------
Temp: 87.52
RPM: 4471.0
Vibration: 6.69
AI Score: 4.6991
STATUS: NORMAL

EDGE AI DETECTOR
------------------------
Temp: 48.04
RPM: 3609.0
Vibration: 9.83
AI Score: 4.7873
STATUS: FAILURE RISK
```

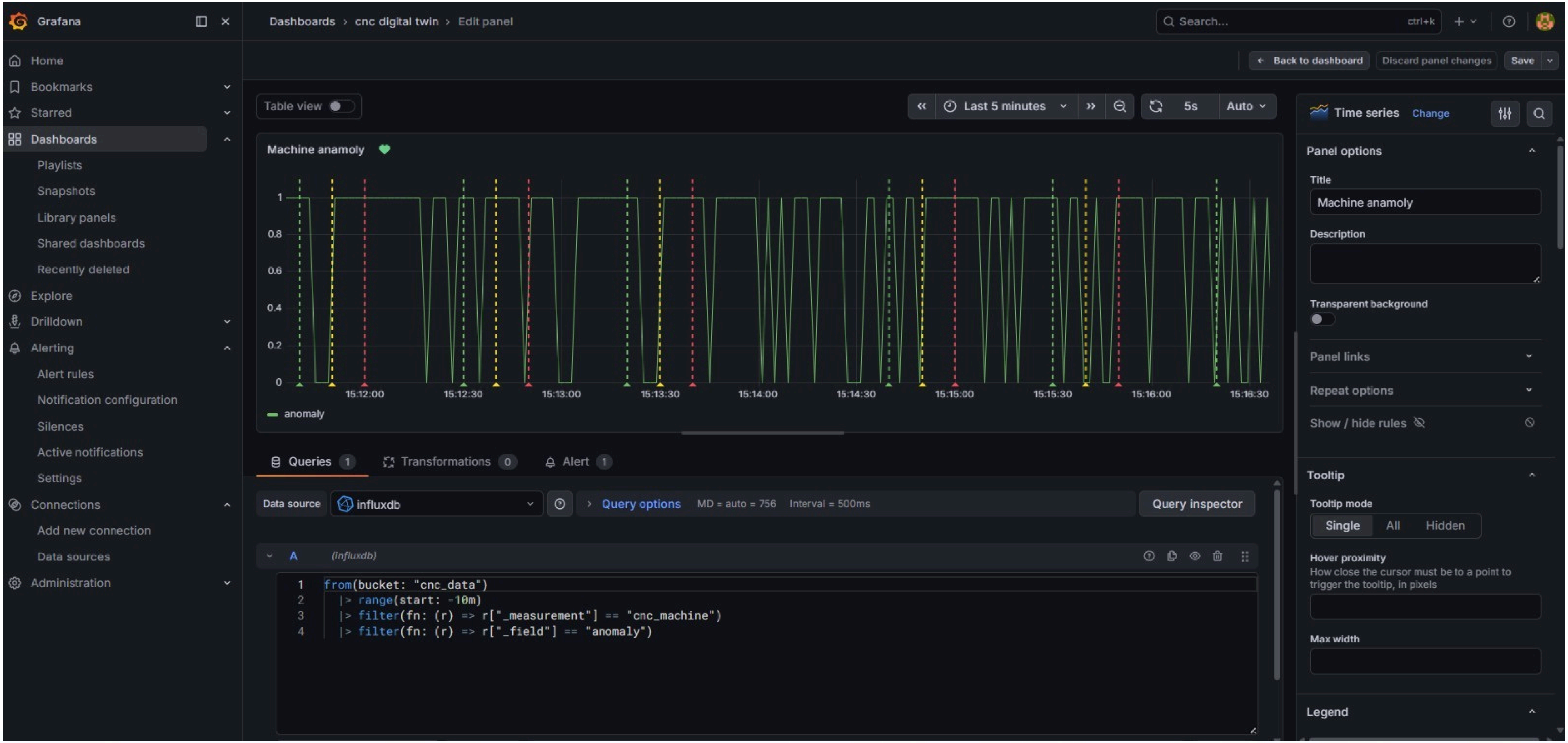
RESULTS AND OUTPUT



RESULTS AND OUTPUT



RESULTS AND OUTPUT



RESULTS AND OUTPUT

RESULTS AND OUTPUT

RESULTS AND OUTPUT

RESULTS AND OUTPUT

RESULTS AND OUTPUT

📁 cnc rpm alert › DatasourceNoData

🔥 1 firing instances

Firing | DatasourceNoData | View alert

Summary

CNC Machine RPM Alert

Description

RPM exceeded safe limit.
Possible machine overload detected.

Labels

alertname	DatasourceNoData
datasource_uid	ffj31ar8jmpz4e
grafana_folder	cnc rpm alert
ref_id	A
rulename	Machine rpm

Silence | View dashboard | View panel

FUTURE ENHANCEMENTS

- Connect real vibration sensors
- Mobile app monitoring
- Cloud analytics integration
- Advanced deep learning models
- Automatic maintenance scheduling

CONCLUSION

- The proposed Digital Twin-based predictive maintenance system provides an efficient solution for CNC machine monitoring.
- By combining vibration sensor fusion, LSTM anomaly forecasting, and edge processing, the system predicts faults early and reduces maintenance downtime.
- It supports smart manufacturing and Industry 4.0 applications

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